

Mirror-Transcending Aerial Imaging System for Multiple Users

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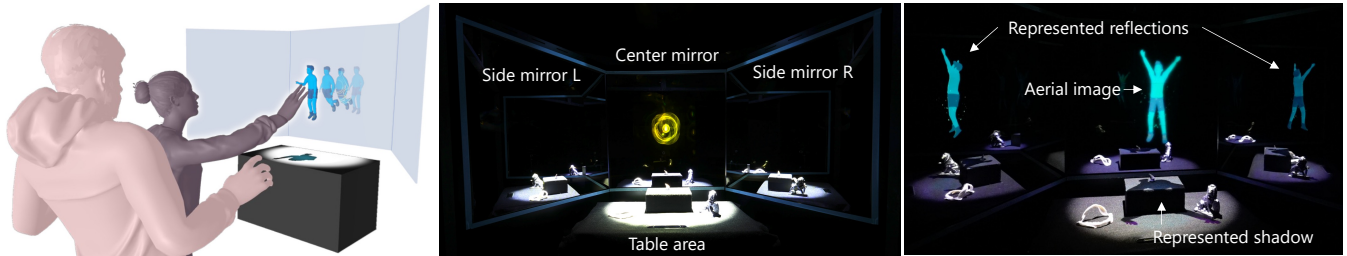


Figure 1: (Left) Experience image. (Middle) Prototype appearance. (Right) Aerial image shown in the physical space.

ABSTRACT

We present a system that enables multiple users to share the viewing experience of mirror-transcending aerial imaging, in which virtual objects continuously move between mirrored and physical spaces. The system consists of an aerial-imaging optical system with a wide viewing range and a control system that synchronizes multiple display devices' position and image rendering. We implemented a prototype to display the mirror-transcending aerial image for multiple users. The prototype looks like a three-sided mirror, allowing four to five users to participate in the viewing experience and share their impressions.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality; Displays and imagers.

KEYWORDS

Interactive mirror, Aerial imaging, Multiple user interface

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1 INTRODUCTION

As part of our ongoing quest to expand our perceptions of the real world, we are exploring ways to achieve a fictional phenomenon created by the human imagination in the real world. To this end, we have focused on mirrors appearing in many works of fiction and developed Mirror-Transcending Aerial Imaging (MiTAI) technology that enables virtual objects to move continuously between mirrors and physical spaces. Our previous study [Makiguchi et al. 2023] proposed and implemented an interactive MiTAI system that uses simple retroreflective aerial imaging optics and investigated content presentation and interaction methods unique to MiTAI. While this system allows users to view aerial images that can move between mirrored and physical spaces, viewers are limited to one at a time due to the narrow viewing area.

By sharing reactions such as eye contact and facial expressions while multiple users simultaneously view MiTAI's fiction-like phenomenon, we expect to strengthen the impression of the experience through empathy, such as the feeling of amazement. Therefore, in the current work, we propose a system that can simultaneously provide multiple users with the viewing experience of MiTAI. We implemented a three-sided mirror prototype that allows four to five users to view aerial images moving in and out of the center mirror. The prototype has an intuitive interaction function between the user and the aerial image. In addition, it reproduces reflections in the side mirrors and a shadow on the table as unique content presentation methods to enhance the presence of aerial images in the physical space.

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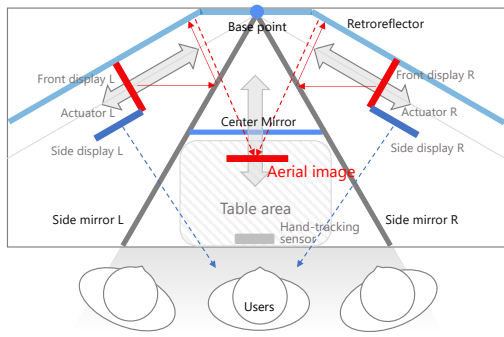


Figure 2: Top view of proposed optical system.

2 METHODS

The proposed system consists of an optical system for forming aerial images and a control system for rendering images and moving displays (Fig. 2).

The optical system comprises three mirrors, four light source displays, retroreflectors, and a projector. The arrangement of the side mirrors and the retroreflectors is based on our previous study [Hoshi et al. 2023], which extended the viewing area for retroreflective aerial imaging optics. We added a center mirror to provide properties that allow the aerial image to transcend the mirror surface. The mirrors are reflective polarizers that reflect one linear polarization direction and transmit the other direction. The transmission axes of all mirrors are in the same direction. The front displays are the light sources for the aerial image. Assuming a base point that couples the side mirrors, the distance between the aerial image and the base point corresponds to the distance between the front display and the base point. Therefore, the depth of the aerial images can be changed by physically moving the front displays.

The side displays are fixed to each front display in an L-shape and moved together. The direction of the polarization of the light emitted by the displays coincides with the mirrors' transmission axis. Therefore, the users can view the image on the side display directly through the mirror. The side displays show the reflected image of the aerial image when displayed in the physical space. We define the user-side area surrounded by three mirrors as the table area. A projector is installed directly above the table area to illuminate the table, direct the surface texture, and show the shadow of the aerial image displayed in the physical space. The right photo in Fig. 1 shows the aerial image in the physical space and how the reflected images and the shadow are reproduced.

The proposed optical system requires time synchronization between two actuator stage positions and five displayed images, including the projector. As the control system, we connected one master PC to three client PCs and a microcontroller (Arduino) via a local network. Each client PC outputs images to two displays or the projector. The applications on the master PC and each client PC have the same virtual space and display different images by switching virtual cameras. The master PC sends synchronization signals to the client PCs using the User Datagram Protocol (UDP) regarding content playback and controls the timeline of the client PCs. In addition, the master PC sends the position of the aerial image in the virtual space to Arduino to synchronize the actuator stage positions.

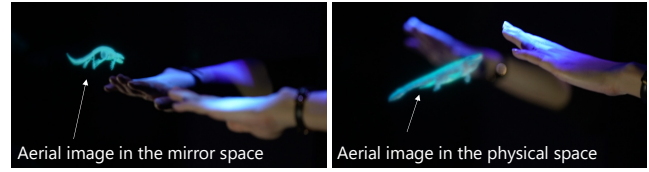


Figure 3: Interaction with aerial image using the hand.

The hand-tracking sensor (UltraLeap 3Di) connected to the master PC detects the user's palm, and the master PC sends its position coordinates to each client PC. Suppose the aerial images are displayed in the mirror space. In that case, the position coordinate of the palm reflected in the mirror space is sent by inverting the coordinate in the depth direction. Through this method, the user can touch the aerial image in the mirror with the palm reflected by the mirror and directly touch the image outside the mirror. This interaction method was initially proposed in our previous study [Makiguchi et al. 2023], and its introduction into the system here allows surrounding users to enjoy watching the interaction between the user and the aerial image.

We implemented a prototype with a center mirror 600 mm in width and height and side mirrors 1200 mm in width and 600 mm in height. The angle of the side mirrors is 60 degrees, and the width of the opening on the user side of the side mirrors is 1200 mm. We used a 15-inch Organic Light Emitting Diode (OLED) monitor in portrait orientation for the displays. The aerial image can move 200 mm outside the mirror and 200 mm inside reference to the center mirror. Under an assumed viewing distance of 1000 mm from the center mirror, the calculated horizontal viewing range is about 2000 mm wide. Therefore, four to five users can watch simultaneously in front of the three-sided mirror.

3 EXPERIENCE

We implemented content inspired by museum exhibits. We placed a dinosaur (Mosasaurus) tooth fossil as a physical object in the table area, and the content begins with a virtual agent describing the fossil. After the description, a skeletal specimen appears as an aerial image on top of the fossil in the center mirror. The aerial image pops out from the mirror when the user touches the aerial image with the palm reflected by the mirror. When the aerial image emerges into the physical space, the image transforms from a skeleton into a skin model and begins to swim as if it were alive again. The aerial image in mirror space faces the direction of the palm reflected in the mirror, whereas the image in physical space faces the direction of the palm in physical space (Fig. 3). This interaction allows the users to observe the aerial image from their desired angle.

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